

RYAN NOWICKI

Astrophysics Ph.D. Candidate at Vanderbilt University

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Curriculum Vitae – Last Updated: May 14th, 2026

Education

Vanderbilt University

Doctor of Philosophy in Astrophysics

Advisor: Karan Jani, Ph.D.

Aug. 2023 – Expected 2028

Nashville, Tennessee

The University of Texas at Austin

Bachelor of Science in Astronomy

Bachelor of Science in Physics

Creative Writing Certificate, Poetry Focus

Sep. 2019 – May 2023

Austin, Texas

Publications

For a full list of my papers including collaboration papers (6 published, 20 preprint), see: [NASA ADS](#) or [InspireHEP](#)

- [1] Tintin Nguyen et al. “Black Hole Binary Detection Landscape for the Laser Interferometer Lunar Antenna (LILA): Signal-to-Noise Calculations & Science Cases”. In: (May 2026). arXiv: [2605.11283 \[astro-ph.HE\]](#).
- [2] H. L. Iglesias et al. “Eccentricity Estimation for Five Binary Black Hole Mergers with Higher-order Gravitational-wave Modes”. In: *The Astrophysical Journal* 972.1 (Aug. 2024), p. 65. DOI: [10.3847/1538-4357/ad5ff6](#). URL: <https://dx.doi.org/10.3847/1538-4357/ad5ff6>.
- [3] Deborah Ferguson et al. “Mayawaves: Python Library for Interacting with the Einstein Toolkit and the MAYA Catalog”. In: *Journal of Open Source Software* 9.98 (June 2024), p. 6032. DOI: [10.21105/joss.06032](#). URL: <https://doi.org/10.21105/joss.06032>.
- [4] Deborah Ferguson et al. “Second MAYA Catalog of Binary Black Hole Numerical Relativity Waveforms”. In: *arXiv e-prints*, arXiv:2309.00262 (Sept. 2023), arXiv:2309.00262. DOI: [10.48550/arXiv.2309.00262](#). arXiv: [2309.00262 \[gr-qc\]](#).

Posters and Presentations

- [1] Ryan Nowicki et al. *Probing the Formation Channels of lite-IMBHs with LISA-driven Multi-band Inference*. **Poster** presented at the Gravitational Wave Physics and Astronomy Workshop. Dec. 2025.
- [2] Ryan Nowicki and Karan Jani. “Do a Flip!”: How LISA will be able to Observe Transitional Precession in Intermediate-Mass Black Hole Binaries. **Poster** presented at Taking Spin Measurements for a Spin: Recent Progress on Black Hole Spin Measurements Across the Electromagnetic and Gravitational Spectra. Sept. 2025.
- [3] Ryan Nowicki and Karan Jani. *Assessing the Detectability of Transitionally Precessing Binary Black Holes in Next-Generation Detectors*. **Talk** given at APS Global Physics Summit. Mar. 2025.
- [4] Ryan Nowicki and Karan Jani. *An Accuracy Test on Approximate Waveforms in Transitionally Precessing Systems*. **Poster** presented at the 15th International LISA Symposium. July 2024. URL: <https://virtual.oxfordabstracts.com/#/event/5189/submission/277>.
- [5] Ryan Nowicki, Jessica Sydnor, and Sarah Burke-Spolaor. “Wrangling a BOBcat: Building a Brand New Database for Binary Black Holes”. In: *American Astronomical Society Meeting Abstracts*. Vol. 55. American Astronomical Society Meeting Abstracts. Jan. 2023, 472.02, p. 472.02.

Professional Affiliations

Establishing Multi-messenger-astronomy Interdisciplinary Training (EMIT) – *Student*

2024 – Present

LISA Consortium – *Core Member*

2025 – Present

Associate Member

2024 – 2025

LIGO Scientific Collaboration – *Member*

2021 – Present

Awards and Honors

Russell G. Hamilton Scholar and University Graduate Fellow <i>Three-year Fellowship at Vanderbilt University</i>	Aug. 2023 – Present <i>Nashville, Tennessee</i>
Harold Stirling Vanderbilt Award <i>Two-year extension of University Graduate Fellowship at Vanderbilt University</i>	May 2026 – Present <i>Nashville, Tennessee</i>
Vanderbilt Physics and Astronomy Research Award <i>Funding granted in support of summer research activities</i>	May 2026 <i>Nashville, Tennessee</i>
Best Poster Prize, Black Hole Spin Workshop, Wake Forest University <i>Poster Prize awarded for poster [2]</i>	Sep. 2025 <i>Winston-Salem, North Carolina</i>
Vanderbilt University Graduate Travel Grant <i>In support of attendance and poster presentation at 15th International LISA Symposium [4]</i>	July 2024, Sep. 2025 <i>Dublin, Ireland</i>
<i>In support of attendance and poster presentation at the Black Hole Spin Workshop [2]</i>	<i>Winston-Salem, North Carolina</i>
Second Prize Poster, Conference for Undergraduate Women in Physics, Texas Christian University <i>Poster Prize awarded for poster [5]</i>	Jan. 2023 <i>Fort Worth, Texas</i>
Nancy Francis and William Arnold McMinn Endowed Presidential Scholarship in Physics <i>Undergraduate Scholarship at the University of Texas at Austin</i>	Aug. 2022 – May 2023 <i>Austin, Texas</i>

Teaching

Vanderbilt University	
• Teaching Assistant; ASTR 1020L - Introductory Daytime Astronomy Lab	Spring 2025
• Teaching Assistant; ASTR 1010L - Introductory Nighttime Astronomy Lab	Spring, Fall 2024
The University of Texas at Austin	
• Learning Assistant; PHY 302K - Mechanics, Heat, and Sound	Fall 2021, Fall 2022, Spring 2023
• Learning Assistant; PHY 317K - General Physics I	Spring 2022
• Learning Assistant; PHY 316 - Electricity and Magnetism	Spring 2021

Outreach and Media

NASA Astronomy Picture of the Day: “Black Holes and Neutron Stars: 218 Mergers and Counting” <i>Scientific graphic created for LIGO and chosen as Astronomy Picture of the Day</i>	Mar. 26, 2026 See Webpage
“Gravitational Wave Transient Catalog: Compact Binary Coalescence Detections...” (poster) <i>Poster presented at the “10-years of Gravitational Waves Vanderbilt Community Celebration”</i>	Sep. 2025 <i>Nashville, Tennessee</i>
Binary Black Hole Dynamics Visualization Tutorial (invited talk) <i>Created and lectured an interactive workshop session at the EMIT Summer School at Vanderbilt</i>	July 2025 <i>Nashville, Tennessee</i>
“How the Flux do we Know Black Holes Exist????” <i>Talk presented at Schrodinger’s Pint, a public physics talks event</i>	Oct. 2022 <i>Austin, Texas</i>